

Exact Unrestricted Bounded-Replacement State Complexity

Rational Binary Balance Languages and the State-Push Staircase

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Abstract

For coprime integers $p, q \geq 1$, let

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The problem studied here originates in a completion-aware 2:1 deterministic pushdown construction and its subsequent $p:q$ generalization, which isolates the signed residual

$$\rho(w) = qN_1(w) - pN_0(w)$$

as the semantic coordinate governing right quotients and completions. A representation-independent one-work-symbol argument gives the exact state cost when stack replacement length is unbounded. If each transition is instead required to replace the visible stack symbol by a word of length at most B , a canonical raw residual encoding gives a natural state staircase as an upper bound, but a priori there is no reason for that bound to be optimal against arbitrary history-dependent pushdown representations.

We prove that it does. Put

$$M = \max(p, q), \quad m = \min(p, q).$$

For every $B \geq 2$, among deterministic right-endmarker ε -pushdown recognizers with a permanent bottom marker and exactly one non-bottom work symbol,

$$Q_{\text{unrestricted, 1work}}^{(B)}(p, q) = M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

The lower bound is representation-independent. Its new ingredient is an *outward first-hit band*: from M distinct semantic phases, the first crossing of a high unary-stack threshold must occur at one of only $B-1$ possible heights. Pigeonhole therefore forces at least $\lceil M/(B-1) \rceil$ phases to first hit one common height, and residual rigidity plus coprimality forces their control states to be distinct. A finite-depth descent synchronizes this outward family with the protected phase family of the opposite residual sign.

The matching upper construction is ε -free and is given explicitly. Consequently the entire one-work-symbol state-push staircase is intrinsic to the unrestricted recognizer class. In particular

$$Q_{\text{unrestricted, 1work}}^{(2)} = 2M, \quad B_{\text{crit}} = 1 + \left\lceil \frac{M}{m} \right\rceil,$$

and the latter is the exact replacement threshold at which the unbounded one-work-symbol optimum $M + m = p + q$ first becomes attainable.

Keywords. deterministic pushdown automata; rational balance languages; bounded replacement; state complexity; descriptive complexity; representation-independent lower bounds.

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1 Introduction

The rational binary balance languages

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p, q) = 1,$$

form a particularly transparent test family. The coprimality assumption is without loss of generality, since dividing p and q by their common divisor does not change the language; it will also be the arithmetic input that separates the semantic phases below. They are useful for separating semantic information from its physical pushdown representation. Membership is controlled by one integer quantity,

$$\rho(w) = qN_1(w) - pN_0(w),$$

yet a deterministic pushdown recognizer may encode that quantity through finite control, current stack height, the zero/nonzero top mode, and a history-dependent choice among physical representatives, with deterministic ε -normalization interleaved between input moves.

The line of work leading to the present question began with a completion-aware 2:1 DPDA. In that machine the stack has an operational interpretation: it records the imbalance that a future completion must cancel. The $p:q$ generalization separates this operational picture from the semantic invariant itself. On residuals the input letters act simply by

$$r \xrightarrow{1} r + q, \quad r \xrightarrow{0} r - p.$$

This viewpoint led first to compact residual realizations and then to representation-independent state/stack lower bounds. The present paper asks what remains true after a different physical constraint is imposed: a single pushdown transition may replace the visible stack symbol by a word of length at most B .

The distinction between the two kinds of statement is essential. In a canonical one-work-symbol residual encoding, bounded replacement can be measured arithmetically: a transition can increase work height by at most $B - 1$, and this produces a sharp state staircase. An unrestricted opponent, however, may use arbitrary history-dependent encodings and arbitrary deterministic ε -moves. The residual need not be recoverable from state or height separately, and there need be no fixed translation-invariant high-stack regime or canonical normalization boundary. Thus a lower bound proved by inspecting a chosen residual encoding does not automatically apply to every recognizer of the language.

A representation-independent one-work-symbol argument gives the baseline

$$|Q| \geq p + q = M + m,$$

whereas the canonical bounded construction developed below gives

$$|Q| \leq M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

The gap is nontrivial precisely when B is small. The purpose of this paper is to close it without reconstructing a canonical residual representation inside an arbitrary competitor.

Main theorem. For every coprime $p, q \geq 1$ and $B \geq 2$,

$$Q_{\text{unrestricted, 1work}}^{(B)}(p, q) = M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right). \quad (1)$$

The upper bound is supplied by an explicit ε -free one-work-symbol construction. The new content is the unrestricted lower bound.

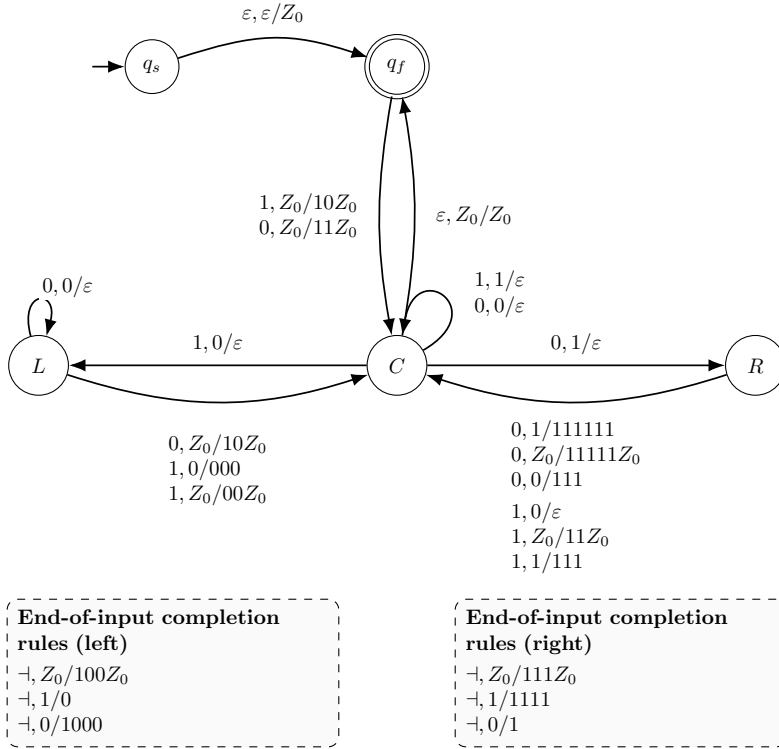


Figure 1: A formalized version of the completion-aware 2:1 DPDA that motivated the residual/completion viewpoint. The center/left/right recognition logic is shown together with the end-of-input completion callouts. The figure records the historical origin only: none of its transition rules, state layout, stack encoding, or completion convention is assumed in the unrestricted bounded-replacement lower bound of the present paper.

New mechanism: first-hit bands. Assume for the moment that $p \geq q$. Choose p nearby negative residual phases and follow each one outward along 0^ω . For a high work-height threshold K , let $E_j(K)$ be the first raw configuration on phase j whose height is at least K . Since one transition can raise height by at most $B - 1$,

$$\text{ht}(E_j(K)) \in \{K, K + 1, \dots, K + B - 2\}.$$

Hence at least

$$\left\lceil \frac{p}{B - 1} \right\rceil$$

phases first hit one common height. Because the stack is unary, equality of control states at that common height would identify the full physical configurations. Residual rigidity then gives a congruence that, by $\gcd(p, q) = 1$, forbids two different phases from sharing a state.

The first-hit family alone gives one signed block of states. The second ingredient is the protected-phase machinery behind the unrestricted one-work-symbol theorem. It supplies p positive phases at arbitrarily high common cuts and, crucially, allows descent by any fixed finite depth. The first-hit band has width $B - 1$, so only depths $0, \dots, B - 2$ need be synchronized. At the resulting common unary stack we obtain simultaneously p positive states and $\lceil p/(B - 1) \rceil$ negative states. Opposite residual signs make the two sets disjoint, giving

$$|Q| \geq p + \left\lceil \frac{p}{B - 1} \right\rceil.$$

Together with $|Q| \geq p + q$, this yields the lower half of (1).

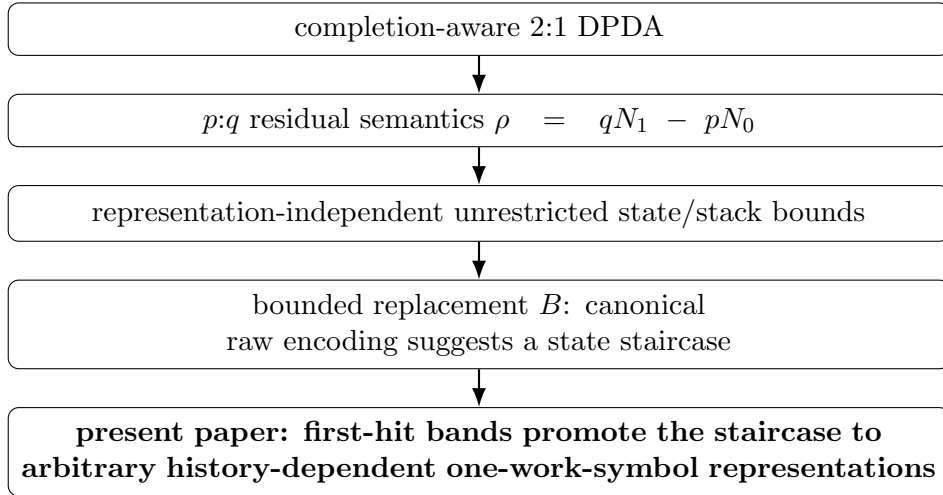


Figure 2: Research progression and the representation gap closed in this paper.

What is and is not claimed. The result is not a new normal-form theorem for pushdown automata in general. Bounded push length can often be traded against auxiliary control in standard simulations. Here the language family and the one-work-symbol physical alphabet are fixed, and the question is the exact parameter-dependent finite-control cost under an arbitrary deterministic pushdown representation. The novelty is that the state–push staircase suggested by the canonical residual construction is proved optimal over the unrestricted recognizer class.

Organization. Section 2 fixes the model and proves residual rigidity. Section 3 gives the matching bounded construction. Section 4 develops the protected unary-stack phase interface and finite-depth descent needed by the new argument. Section 5 proves the outward first-hit-band lemma. Section 6 synchronizes the two signed families and proves the exact theorem. Section 7 derives the unrestricted state–push staircase and its inverse form.

2 Model and semantic rigidity

2.1 Bounded-replacement one-work-symbol pushdown recognizers

A deterministic right-endmarker ε -pushdown automaton has a finite control set Q , binary input alphabet $\{0, 1\}$ followed by one mandatory symbol $\neg$, and stack alphabet

$$\Gamma = \{Z_0, A\},$$

where Z_0 is a permanent bottom marker and A is the unique work symbol; the initial stack is Z_0 . Permanence means that Z_0 is never removed and never occurs above the bottom: a rule reading Z_0 replaces it only by a word of the form Z_0A^t . Acceptance is by final state after the mandatory endmarker has been consumed. The stack top is written on the right. The *work height* of Z_0A^h is h . A raw transition has the form

$$(q, \eta X, av) \vdash (q', \eta \beta, v)$$

or its ε -analogue, where X is the visible top symbol. Determinism means that for each state/top pair there is at most one move on each symbol in $\{0, 1, \neg\}$ and, whenever an ε -move is enabled, no input-consuming move is enabled at that state/top pair.

Fix $B \geq 2$. The machine is *B-bounded* if every replacement word β has length at most B . Thus one raw transition can decrease work height by at most one and can increase it by at most $B - 1$. The opponent may otherwise use arbitrary deterministic ε -moves, noncanonical intermediate configurations, and arbitrary history-dependent representations. A reachable configuration is called *settled* when no forced ε -move is enabled, so the machine is ready either to consume the next input symbol or the right endmarker. When comparing runs on different inputs, we call the pair consisting of the current control state and the complete stack Z_0A^h the *physical configuration*; thus “physically identical” suppresses the unread input suffix but requires equality of both state and complete stack.

The lower bound uses only the binary-prefix behavior of this model and the following liveness consequence: every finite prefix occurring below has a finite binary continuation followed by $\neg$ that is accepted. Hence deterministic ε -divergence cannot permanently prevent the next required input symbol from being consumed.

Definition 2.1 (Unrestricted bounded one-work-symbol state complexity). Let

$$Q_{\text{unrestricted}, 1\text{work}}^{(B)}(p, q)$$

be the minimum $|Q|$ among deterministic B -bounded right-endmarker ε -pushdown recognizers of $L_{p,q}$ with permanent bottom marker and exactly one non-bottom work symbol.

2.2 Residual quotient

For $w \in \{0, 1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then $w \in L_{p,q}$ iff $\rho(w) = 0$.

Lemma 2.2 (Integer residual realizability). *For every $t \in \mathbb{Z}$ there is a word $z \in \{0, 1\}^*$ with $\rho(z) = t$.*

Proof. By Bezout, choose integers x_0, y_0 with $qx_0 - py_0 = t$. Replacing (x_0, y_0) by $(x_0 + pn, y_0 + qn)$ for sufficiently large n makes both entries nonnegative. Then $z = 1^{x_0+pn}0^{y_0+qn}$ has residual t . $\square$

Lemma 2.3 (Residuals classify right quotients). *For binary prefixes u, v ,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z ,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 2.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ but $vz \notin L_{p,q}$. $\square$

Lemma 2.4 (Physical residual rigidity). *If two reachable raw configurations of a deterministic recognizer are physically identical, then the consumed binary prefixes have the same residual.*

Proof. From one physical configuration, determinism gives the same future behavior on every finite continuation. Hence the two consumed prefixes have equal right quotients. Apply Theorem 2.3. $\square$

It is therefore legitimate to write $\text{Res}(C)$ for the unique residual carried by a reachable raw configuration C . Input-consuming transitions satisfy

$$\text{Res}(C \xrightarrow{1} C') = \text{Res}(C) + q, \quad \text{Res}(C \xrightarrow{0} C') = \text{Res}(C) - p,$$

while ε -moves preserve Res .

3 A matching bounded one-work-symbol construction

By symmetry it is enough to assume $p \geq q$. Put

$$b = b_B := \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

Use positive states $P_0, \dots, P_{p-1}$ and negative states $N_0, \dots, N_{b-1}$. Encode residuals as follows. For $r \geq 0$, write

$$r = pk + s, \quad k \geq 0, \quad 0 \leq s < p,$$

and set

$$\eta(r) = (P_s, Z_0 A^k).$$

For $r < 0$, write uniquely

$$r = u - bh, \quad 0 \leq u < b, \quad h \geq 1,$$

and set

$$\eta(r) = (N_u, Z_0 A^{h-1}).$$

Proposition 3.1 (Bounded residual construction). *For $p \geq q$ and $B \geq 2$, the encoding above is realized by an ε -free deterministic B -bounded one-work-symbol pushdown automaton with exactly*

$$p + b_B$$

states. Hence, for arbitrary coprime p, q ,

$$Q_{\text{unrestricted, 1work}}^{(B)}(p, q) \leq M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

Proof. Take (P_0, Z_0) as the initial configuration. We verify that the two residual translations can be implemented by one top rewrite.

On the positive side, input 0 sends $pk + s$ to $p(k - 1) + s$ when $k \geq 1$, so one A is popped. At the boundary $k = 0$, write

$$s - p = u - bh, \quad 0 \leq u < b, \quad h = \left\lceil \frac{p - s}{b} \right\rceil.$$

Since $p \leq b(B - 1)$, one has $h \leq B - 1$; replacing Z_0 by $Z_0 A^{h-1}$ has length at most B .

For input 1, put

$$s' = (s + q) \bmod p, \quad c = \frac{s + q - s'}{p} \in \{0, 1\}.$$

Then $pk + s + q = p(k + c) + s'$. Thus the stack height changes by at most one, implemented either by $A \mapsto A^{1+c}$ or by the corresponding bottom-marker rule.

On the negative side, write $r = u - bh$. For input 0, let

$$u' = (u - p) \bmod b, \quad d = \frac{u' - u + p}{b} = \left\lceil \frac{p - u}{b} \right\rceil.$$

Then

$$r - p = u' - b(h + d).$$

Again $d \leq B - 1$, so an existing top A is rewritten as A^{d+1} , or at height zero the bottom marker is rewritten as $Z_0 A^d$; both replacement words have length at most B .

Finally, because $b \geq q$, input 1 changes the negative quotient height by at most one. If $u + q < b$, the height is unchanged and the remainder becomes $u + q$. If $u + q \geq b$ and $h \geq 2$, one work symbol is popped and the remainder becomes $u + q - b$. If $u + q \geq b$ and $h = 1$, then

$$0 \leq u + q - b < q \leq p,$$

so the transition crosses directly to the positive state P_{u+q-b} while leaving the stack Z_0 unchanged. All binary rules are therefore deterministic, ε -free, and B -bounded, and they realize $r \mapsto r - p$ and $r \mapsto r + q$ exactly. Finally designate P_0 as the sole final state and, among right-endmarker moves, define only

$$\delta(P_0, Z_0, \dashv) = (P_0, Z_0).$$

Since $\eta(0) = (P_0, Z_0)$ and no nonzero residual representative has both state P_0 and top Z_0 , the mandatory endmarker accepts exactly residual zero without introducing another state. $\square$

4 Protected unary-stack phases and finite-depth descent

The first-hit block constructed later must be compared with a representation-independent family of opposite-sign phases living over one common unary stack. We therefore isolate the protected unary interface needed for the synchronization argument. All statements in this section concern an arbitrary deterministic recognizer in the model of Section 2; no bounded replacement property is used until Section 5.

Consider the live raw computation on 1^ω . Every finite prefix 1^n has a finite accepting extension by Theorem 2.2. Hence the run consumes arbitrarily many input symbols and cannot become trapped in a deterministic ε -cycle before the next required 1 is read.

Lemma 4.1 (Unary-stack height escape). *Along the raw computation on 1^ω , work height eventually leaves every finite bound.*

Proof. No full raw state/stack configuration can recur. A recurrence after different numbers of consumed 1's would assign the same physical configuration two different residuals by Theorem 2.4; a recurrence at the same input position would create a deterministic ε -cycle and prevent the next live input symbol from being read. If work height failed to leave some finite bound, finitely many control states and unary stacks would force such a recurrence. $\square$

Because the integer-valued height tends to infinity in the preceding sense, there are suffix minima at arbitrarily large heights. At a suffix minimum, the remainder of the 1^ω run never goes below the current complete unary stack.

Lemma 4.2 (Protected unary-stack pump). *There exist a state r , integers $a, c, u \geq 1$, and an infinite family of reachable configurations*

$$(r, Z_0 A^{h_k}), \quad h_k = a + uk,$$

such that one fixed raw macro consuming exactly c symbols 1 maps

$$(r, Z_0 A^{h_k}) \xrightarrow{1^c} (r, Z_0 A^{h_k+u})$$

for every $k \geq 0$. The macro never exposes a source cell below height h_k and is therefore independent of the buried unary context.

Proof. Among suffix minima of arbitrarily large height, some state/top mode occurs twice, the later occurrence at a strictly larger height. Between the two occurrences the computation never drops below the earlier suffix minimum, so the buried stack below that cut is not inspected. Since the work alphabet is unary, the later stack differs from the earlier one by $u \geq 1$ additional copies of A . The raw transition sequence is therefore valid over every copy of the same buried context and can be iterated. It must consume at least one input symbol: otherwise a deterministic height-increasing ε -macro from one mode back to itself would iterate forever, contradicting liveness. Writing the first chosen height as a gives $h_k = a + uk$. $\square$

The next lemma is the unary form of finite-depth exposure. It is the point at which arbitrary history-dependent representations are neutralized: a fixed finite layer above a protected cut cannot remain hidden forever. We say that a family of configurations is obtained from the protected family by one *fixed relative history above height h_k* if one finite raw transition pattern, with the same sequence of input-consuming and ε -moves and the same rewrite rules, translates over every cut $Z_0 A^{h_k}$, never visits height below h_k , and ends in the stated common state and fixed height offset. Because all symbols above a positive cut are A , such a history is independent of the buried unary context.

Lemma 4.3 (Finite-depth unary-stack descent). *Fix $d, D \geq 0$ and a reachable family*

$$(q, Z_0 A^{h_k+d}), \quad h_k = a + uk \longrightarrow \infty,$$

obtained from the protected family by one fixed relative history above height h_k . Then, after discarding finitely many k so that $h_k \geq D$, there are an integer $e_{d,D} \geq 0$ and a state $q_{d,D}$, independent of k , such that following 0^ω until the first exposure of the shifted cut at height $h_k - D$ gives

$$(q, Z_0 A^{h_k+d}) \xrightarrow{0^{e_{d,D}}} (q_{d,D}, Z_0 A^{h_k-D}).$$

The displayed macro includes all forced ε -moves. In particular, $D = 0$ gives descent to the original protected cut.

Proof. Put $t_k = h_k - D$. Before the shifted cut at height t_k is exposed, the relative configuration above it starts from the fixed pair (q, A^{d+D}) and follows a deterministic raw history independent of k . Hence either this relative stack is emptied after one fixed finite history for every sufficiently large k , or it is never emptied for any sufficiently large k . Because one raw transition decreases work height by at most one, first exposure of the cut occurs at height exactly t_k .

Assume the second alternative. The relative 0^ω computation is live. Raw recurrence is impossible by Theorem 2.4, so its relative height escapes. Applying the suffix-minimum argument of Theorem 4.2 with 0 in place of 1 to this relative computation produces a second context-independent pump, now consuming $e \geq 1$ zeros and adding $v \geq 1$ work cells. After a fixed prefix, variation of the original protected-family parameter k and independent iteration of the new 0-pump yield reachable configurations in one fixed control state with residual

$$C + kcq - \ell ep$$

and complete unary height

$$H + uk + v\ell$$

for all sufficiently large k, ℓ , where the constants C, H absorb the fixed relative history and the shift D .

Compare the parameter pairs $(k + v, \ell)$ and $(k, \ell + u)$. Their unary heights, hence their complete stacks, are identical, and their control states are identical; but their residuals differ by

$$vcq + uep > 0.$$

This contradicts physical residual rigidity. Permanent nonexposure is therefore impossible, so the shifted cut is reached after one fixed relative history. The number $e_{d,D}$ of consumed zeros and the exposure state $q_{d,D}$ are consequently independent of k . $\square$

We now extract the p positive residual phases used later. Starting at $(r, Z_0 A^{h_k})$, consume $j \in \{0, \dots, p-1\}$ further symbols 1 and all forced ε -moves. Since the future on 1^ω is obtained by iterating the same protected macro, the computation above the cut is a fixed relative history for each j ; thus for some state q_j and integer $d_j \geq 0$, independent of k , the resulting configuration is

$$(q_j, Z_0 A^{h_k + d_j}).$$

Apply Theorem 4.3 with $(d, D) = (d_j, 0)$.

Proposition 4.4 (Protected positive phases and simultaneous descent). *There are affine protected heights $h_k = a + uk$ and p reachable configurations*

$$(s_j, Z_0 A^{h_k}), \quad j = 0, \dots, p-1,$$

whose states $s_0, \dots, s_{p-1}$ are pairwise distinct. Their residuals tend to $+\infty$ as $k \rightarrow \infty$.

Moreover, for every fixed $D \geq 0$, after discarding finitely many k there are states $s_{j,D}$ and integers $z_{j,D} \geq 0$, independent of k , such that

$$(s_j, Z_0 A^{h_k}) \xrightarrow{0^{z_{j,D}}} (s_{j,D}, Z_0 A^{h_k - D}),$$

and the p states $s_{j,D}$ are pairwise distinct. For each fixed D , the descended residuals also tend to $+\infty$ with k . The conclusion holds simultaneously for all D in any prescribed finite set.

Proof. The first family is obtained from the preceding construction by descending the fixed upper depth d_j for each phase. If phases $j \neq t$ exposed the common stack $Z_0 A^{h_k}$ in the same state, and if z_j, z_t are the corresponding numbers of consumed zeros, then residual rigidity would give

$$q(j - t) = p(z_j - z_t).$$

Coprimality forces $p \mid (j - t)$, impossible for $0 < |j - t| < p$. Hence the exposure states are pairwise distinct. The protected pump contributes kcq to every phase residual, so all phase residuals tend to $+\infty$.

For a further fixed depth D , each exposed phase family is still obtained from the original protected family by one fixed relative history ending at height h_k . Apply Theorem 4.3 with $(d, D) = (0, D)$ to each of the p exposed phase families. Equality of two descended states at the common stack would again make their residuals equal. The residual difference between phases j and t always has the form

$$q(j - t) - pn$$

for some integer n , namely the difference of the total numbers of consumed zeros along the two phase histories. Equality would therefore force $q(j - t) = pn$, impossible by coprimality and $0 < |j - t| < p$. Thus phase separation survives descent. Since the additional zero history for a fixed D is independent of k , the common kcq term still drives every descended residual to $+\infty$. For a finite set of depths, take the maximum of the finitely many eventuality thresholds; all required descents then hold simultaneously. $\square$

By exchanging 0 with 1 and p with q , one obtains a symmetric protected negative family with q phases and affine protected heights. We use that symmetric family only in the baseline $p + q$ theorem; the new bounded-replacement argument will construct its own negative block by first-hit geometry.

5 Outward first-hit bands

Assume $p \geq q$. Such settled configurations exist at arbitrarily negative residuals: start from a sufficiently long prefix 0^n and follow its forced ε -moves to the next settled configuration, which must be reached by liveness. Choose one whose residual R is sufficiently negative that

$$R + (p - 1)q < 0.$$

For $j = 0, \dots, p - 1$, consume 1^j and all forced ε -moves until the machine is again ready to inspect the next binary input symbol. Denote the resulting reachable configuration by C_j . Its residual is

$$\text{Res}(C_j) = R + jq < 0.$$

From every C_j follow the unique live computation on 0^ω , including all forced ε -moves. After n consumed zeros the residual is

$$R + jq - np.$$

The settled heights on each such ray are unbounded: otherwise finitely many states and bounded unary heights would force a repeated physical configuration carrying two different residuals.

Lemma 5.1 (Outward first-hit band). *Let K exceed the heights of $C_0, \dots, C_{p-1}$. For each j , let $E_j(K)$ be the first raw configuration on the outward 0^ω run from C_j whose work height is at least K . Then*

$$\text{ht}(E_j(K)) \in \{K, K + 1, \dots, K + B - 2\}.$$

Consequently there is a subset $J_K \subseteq \{0, \dots, p - 1\}$ with

$$|J_K| \geq \left\lceil \frac{p}{B - 1} \right\rceil$$

such that all configurations $E_j(K)$, $j \in J_K$, have one common complete unary stack height. Their control states are pairwise distinct.

Proof. The first-hit configuration exists because the settled heights on the ray are unbounded. The raw configuration immediately preceding it has work height at most $K - 1$. One B -bounded top rewrite can increase work height by at most $B - 1$, so the first-hit height is at most $K + B - 2$. This proves the band statement.

There are p phases and only $B - 1$ possible first-hit heights, so pigeonhole gives the stated subset. Suppose two distinct indices $j, t \in J_K$ have the same control state. Since their stacks are unary and have the same height, the two full physical configurations are identical. Let z_j, z_t be the numbers of consumed zeros at the respective first-hit positions. Residual rigidity gives

$$R + jq - pz_j = R + tq - pz_t,$$

so

$$q(j - t) = p(z_j - z_t).$$

Because $\gcd(p, q) = 1$, one has $p \mid (j - t)$, impossible for $0 < |j - t| < p$. Hence the states are pairwise distinct. $\square$

Remark 5.2. *The lemma uses the replacement bound only through the finite overshoot band of width $B - 1$. It makes no assumption about a residual section, a homogeneous tail, the timing of ε -moves, or the history by which the current unary height was created.*

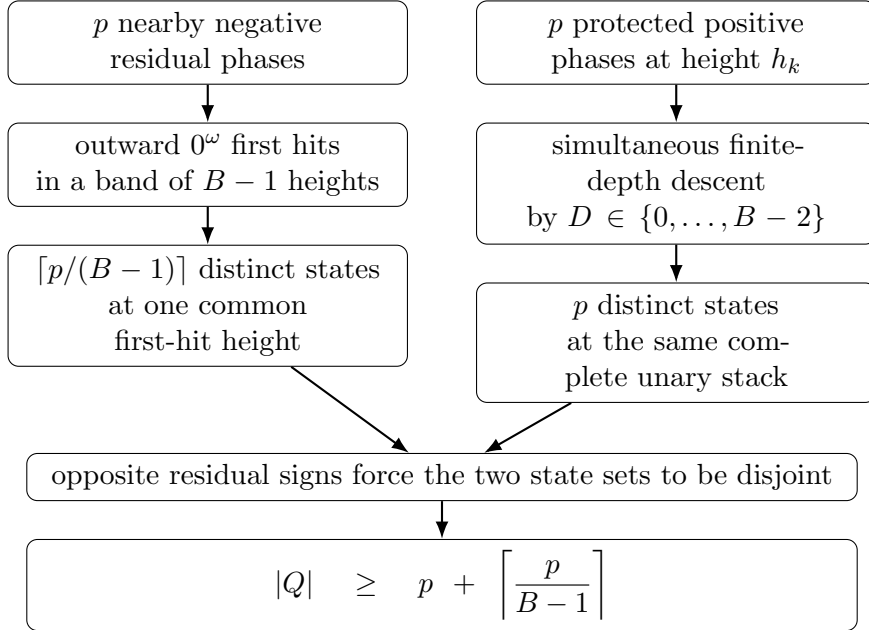


Figure 3: Architecture of the new unrestricted bounded-replacement lower bound for $p \geq q$. The bounded rewrite length is used only to create the finite first-hit band; all representation freedom before that crossing is absorbed by residual rigidity and protected finite-depth descent.

6 Synchronization and the exact theorem

We now synchronize the negative first-hit block with the positive protected family at one complete physical stack.

Theorem 6.1 (First-hit/protected-phase synchronization). *Assume $p \geq q$. Every unrestricted deterministic B -bounded one-work-symbol recognizer of $L_{p,q}$ satisfies*

$$|Q| \geq p + \left\lceil \frac{p}{B - 1} \right\rceil.$$

Proof. Take arbitrarily high positive protected heights h_k from Theorem 4.4. Put

$$K_k = h_k - (B - 2).$$

For k large, K_k lies above all initial negative phase heights C_j , so Theorem 5.1 applies. Its common first-hit height has the form

$$K_k + d_k = h_k - D_k, \quad 0 \leq d_k \leq B - 2, \quad D_k = (B - 2) - d_k \in \{0, \dots, B - 2\}.$$

By the simultaneous finite-depth form of Theorem 4.4, once k is large enough the positive protected family can be descended by whichever D_k occurs. Since only the finitely many depths $0, \dots, B - 2$ are involved, enlarge k once more so that all descended phase residuals at all these depths are positive. At the common complete stack

$$Z_0 A^{h_k - D_k}$$

there are therefore

- (i) p reachable positive-residual configurations in pairwise distinct states, and
- (ii) at least $\lceil p/(B - 1) \rceil$ reachable negative-residual first-hit configurations in pairwise distinct states.

The two state sets are disjoint. Indeed, equality of a positive state and a negative state at the same complete unary stack would identify two reachable physical configurations carrying residuals of opposite sign, contradicting Theorem 2.4. The displayed lower bound follows. $\square$

We also use the representation-independent one-work-symbol bound.

Theorem 6.2 (Unrestricted one-work-symbol baseline). *Every deterministic right-endmarker ε -pushdown recognizer of $L_{p,q}$ with a permanent bottom marker and one work symbol satisfies*

$$\boxed{|Q| \geq p + q.}$$

Proof. Let the positive and negative protected cut heights be

$$h_k^+ = a + uk, \quad h_\ell^- = b + v\ell, \quad u, v \geq 1.$$

Put $g = \gcd(u, v)$. Choose sufficiently large indices k_0, ℓ_0 and define

$$k_n = k_0 + \frac{v}{g}n, \quad \ell_n = \ell_0 + \frac{u}{g}n.$$

Then

$$h_{k_n}^+ - h_{\ell_n}^- = D$$

is independent of n . If $D \geq 0$, apply the finite-depth descent theorem to all p positive phases with depth D and leave the q negative phases at depth zero. For all sufficiently large n , the resulting $p + q$ configurations have the identical complete unary stack

$$Z_0 A^{h_{\ell_n}^-}.$$

The p positive states are pairwise distinct by phase separation, and the symmetric argument separates the q negative states. Their residuals have opposite signs for sufficiently large n , so a positive and a negative configuration cannot share a state at the common stack by Theorem 2.4. Hence $|Q| \geq p + q$. If $D < 0$, descend the negative family by $-D$ instead. $\square$

Theorem 6.3 (Exact unrestricted bounded-replacement one-work-symbol complexity). *For every coprime $p, q \geq 1$ and every $B \geq 2$, with*

$$M = \max(p, q), \quad m = \min(p, q),$$

one has

$$Q_{\text{unrestricted, 1work}}^{(B)}(p, q) = M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

Proof. Assume first $p \geq q$. By Theorem 6.1,

$$|Q| \geq p + \left\lceil \frac{p}{B-1} \right\rceil,$$

while Theorem 6.2 gives $|Q| \geq p + q$. Therefore

$$|Q| \geq p + \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

The case $q > p$ is obtained by symmetry. The matching upper bound is Theorem 3.1. $\square$

7 The unrestricted state–push staircase

Put

$$Q_B := M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

The exact theorem turns the bounded replacement parameter into a completely explicit staircase. The restriction $B \geq 2$ is the first nontrivial regime: with $B = 1$, the initial work height zero can never increase, so only finitely many physical configurations are reachable. By Theorem 2.3, $L_{p,q}$ has infinitely many right quotients and is therefore nonregular; thus $B = 1$ cannot recognize it in this model.

Corollary 7.1 (The first two bounded levels). *For every coprime p, q ,*

$$Q_2 = 2M,$$

and

$$Q_3 = M + \max\left(m, \left\lceil \frac{M}{2} \right\rceil\right).$$

Corollary 7.2 (Critical replacement threshold). *The unbounded one-work-symbol optimum $M + m = p + q$ is attained if and only if*

$$B \geq B_{\text{crit}} := 1 + \left\lceil \frac{M}{m} \right\rceil.$$

Proof. By Theorem 6.3, $Q_B = M + m$ exactly when

$$\left\lceil \frac{M}{B-1} \right\rceil \leq m,$$

which is equivalent to $B - 1 \geq M/m$. $\square$

Corollary 7.3 (Inverse state-to-replacement law). *Fix a state budget $S \geq M + m$. The least replacement bound permitting an unrestricted one-work-symbol recognizer with at most S states is*

$$B_{\min}(S) = \begin{cases} 2, & S \geq 2M, \\ 1 + \left\lceil \frac{M}{S - M} \right\rceil, & M + m \leq S < 2M. \end{cases}$$

Proof. If $S \geq 2M$, Theorem 7.1 shows that $B = 2$ already suffices, and no smaller nontrivial replacement bound exists. Suppose $M + m \leq S < 2M$. Since $S - M \geq m$, Theorem 6.3 gives $Q_B \leq S$ exactly when

$$\left\lceil \frac{M}{B - 1} \right\rceil \leq S - M,$$

which is equivalent to

$$B - 1 \geq \frac{M}{S - M}.$$

The least integer B satisfying this is the displayed value. $\square$

Theorem 7.4 (Exact unrestricted one-work-symbol state-push Pareto staircase). *At fixed total stack alphabet size $|\Gamma| = 2$, the Pareto-minimal pairs $(|Q|, B)$ are exactly*

$$\{(Q_B, B) : 2 \leq B \leq B_{\text{crit}}, B = 2 \text{ or } Q_B < Q_{B-1}\}.$$

Proof. A plateau value is dominated by the same state count with the smaller replacement bound. Every point after B_{crit} is dominated by the critical-threshold point. At every strict drop, reducing B forces a larger state count by Theorem 6.3; hence every displayed breakpoint is Pareto-minimal. $\square$

For the motivating 2:1 case, $M = 2$ and $m = 1$, so

$$Q_B = 2 + \max\left(1, \left\lceil \frac{2}{B - 1} \right\rceil\right).$$

Thus $Q_2 = 4$, while $Q_B = 3$ for every $B \geq 3$. The bounded-push transition from four states to the unrestricted one-work-symbol optimum of three states therefore occurs exactly at $B = 3$.

8 Why the lower bound is genuinely unrestricted

The proof of Theorem 6.3 never assumes that the competitor stores the residual in a chosen state/height coordinate. In particular it uses none of the following:

- raw residual conjugacy;
- a prescribed translation-invariant high-stack regime;
- realtime computation;
- a history-independent representative of a residual;
- a prescribed normalization boundary;
- a canonical relationship between residual magnitude and stack height.

The bounded replacement parameter enters only at the first crossing of a physical height threshold. That crossing has at most $B - 1$ possible overshoots. The semantic phase information is then separated by residual rigidity and coprimality, while the protected family supplies the opposite-sign block at the same complete unary stack. This is why a representation-relative state-push staircase becomes a language-intrinsic unrestricted theorem.

9 Related work and conclusion

State complexity for deterministic pushdown automata has a classical hierarchy theory; see, for example, Harrison’s monograph for the traditional state-hierarchy setting [1]. A different line asks how many pushdown symbols are necessary. Meduna, Vrábel, and Zemek established stack-alphabet hierarchies for stateless pushdown models, and Masopust obtained binary witnesses for the deterministic stateless case and extensions to fixed-state realtime DPDAs [2, 3]. More recent descriptonal-complexity work has studied joint state/stack-symbol measures for pushdown automata [4]. In standard PDA presentations where one transition may replace the top symbol by an arbitrary stack word, the maximum word written in one move is itself part of the description size; Pighizzini and Prigioniero explicitly discuss normalizing such rules to unit pushes with polynomial descriptonal overhead [6]. This standard simulation fact does not determine the minimum number of states when the stack alphabet is fixed and the local replacement budget B is kept as a parameter, which is the tradeoff studied here.

With one non-bottom work symbol, the storage mechanism is closely related to a one-counter device. Deterministic one-counter automata go back at least to Valiant and Paterson [5]. Modern work on pushdown and one-counter memory usage studies whether counter or stack height is constant or how it grows with input length [7]; push complexity instead counts how many push operations are needed along accepting computations [8]. Both resource questions are orthogonal to the one considered here: our stack may grow without a prescribed input-length bound and may use arbitrarily many moves, while the local update magnitude of each raw rewrite is restricted by the parameter B .

The present theorem therefore concerns a different descriptonal axis. We fix the arithmetic language family

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\},$$

fix the physical stack alphabet to one work symbol plus the bottom marker, allow arbitrary deterministic ε -normalization and history-dependent representations, and vary only the maximum replacement length B . Within this model we determine the exact minimum number of control states. Our claim is not that bounded push length itself is a new automata-theoretic resource, nor that arbitrary long pushes cannot be compiled into shorter ones with auxiliary control. The contribution is the exact, representation-independent parameter law for this fixed natural language family.

The conceptual progression is

$\begin{aligned} 2:1 \text{ completion-aware DPDA} &\longrightarrow p:q \text{ residual semantics} \longrightarrow \text{one-work-symbol rigidity} \\ &\longrightarrow \text{bounded first-hit geometry} \longrightarrow \text{exact state-push staircase.} \end{aligned}$
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The first-hit-band mechanism is local in physical height but global in consequence: a bound on one raw rewrite creates a finite overshoot band, pigeonhole concentrates semantic phases at a common complete unary stack, and residual rigidity turns that concentration into finite-control cost. Combined with protected finite-depth descent, this promotes the bounded staircase from a chosen residual encoding to the full unrestricted recognizer class for every $B \geq 2$.

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